Skill 3: Ubuntu EC2 Instance with Nginx Using User Data

Course: Cloud Infrastructure and Services (25CS2206E)

Trimester: 2026-2027 Even Trimester (T5)

Name: Talluri Himabindu Sree

Roll No: 2520030484

Aim

To launch an Ubuntu EC2 instance, install the Nginx web server automatically through a user data script at launch time, open the default page in a browser, and note how this differs from the Amazon Linux + Apache setup.

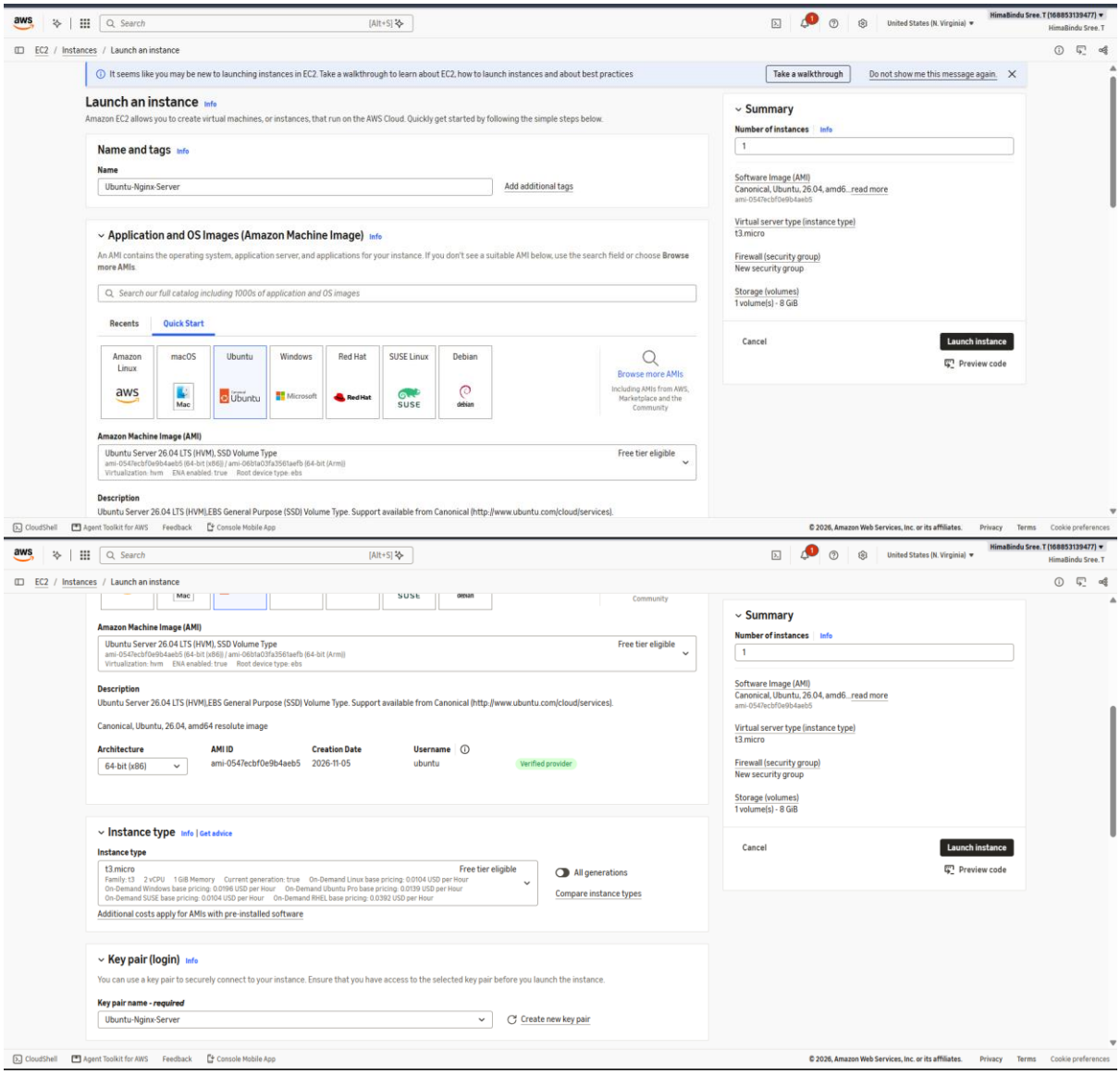
Prerequisites

- An active AWS account with access to the Management Console
- Basic idea of what an EC2 instance is
- Internet connection and a web browser
- An existing key pair (or I can skip SSH, since I only need to check the result in the browser)

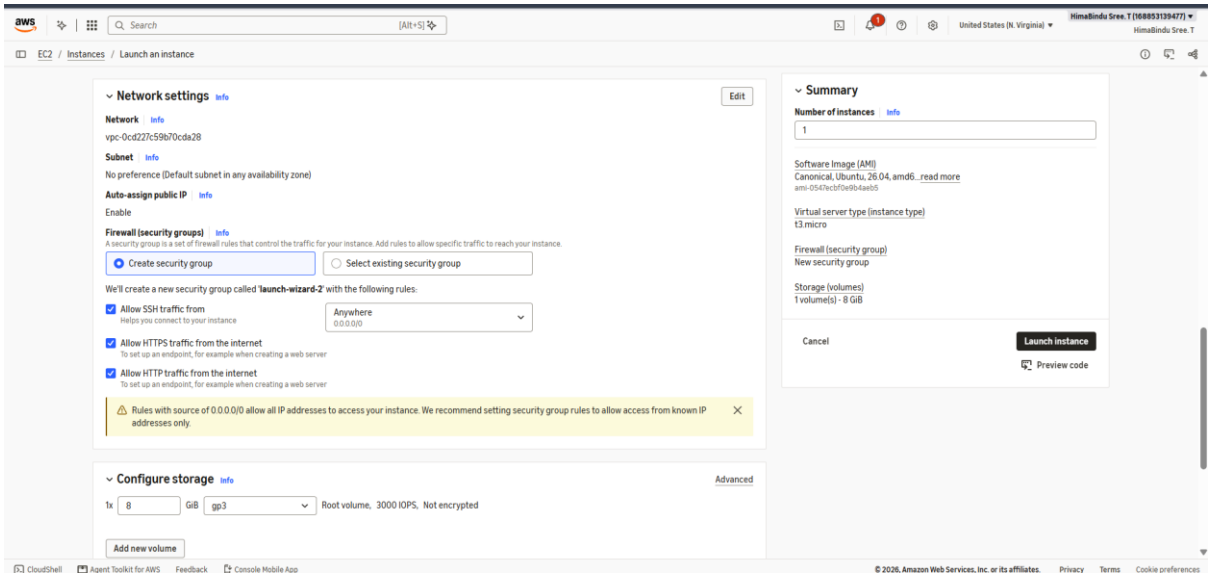
Procedure

Part A: Launching the Ubuntu instance

1. Logged in to the AWS Management Console and searched for EC2 to open the dashboard.
2. Went to Instances in the left menu and clicked Launch instances.
3. Named the instance Ubuntu-Nginx-Server.
4. Under Application and OS Images, picked Ubuntu and chose the Ubuntu Server LTS AMI (26.04 LTS, HVM).
5. Chose a Free Tier eligible instance type, t3.micro.
6. Selected my existing key pair under Key pair (login).
7. In Network settings, allowed SSH traffic and also ticked Allow HTTP traffic from the internet, so the security group opens port 80.
8. Left storage on the default settings.



Screenshot 1: Launch instance page with name, Ubuntu AMI, instance type and key pair



Screenshot 2: Network settings showing SSH and HTTP allowed

Part B: Adding the user data script

1. Expanded Advanced details (everything else left as default) and scrolled to the User data box.
2. Pasted the script below:

```
#!/bin/bash
# Update Ubuntu package repositories
sudo apt update -y

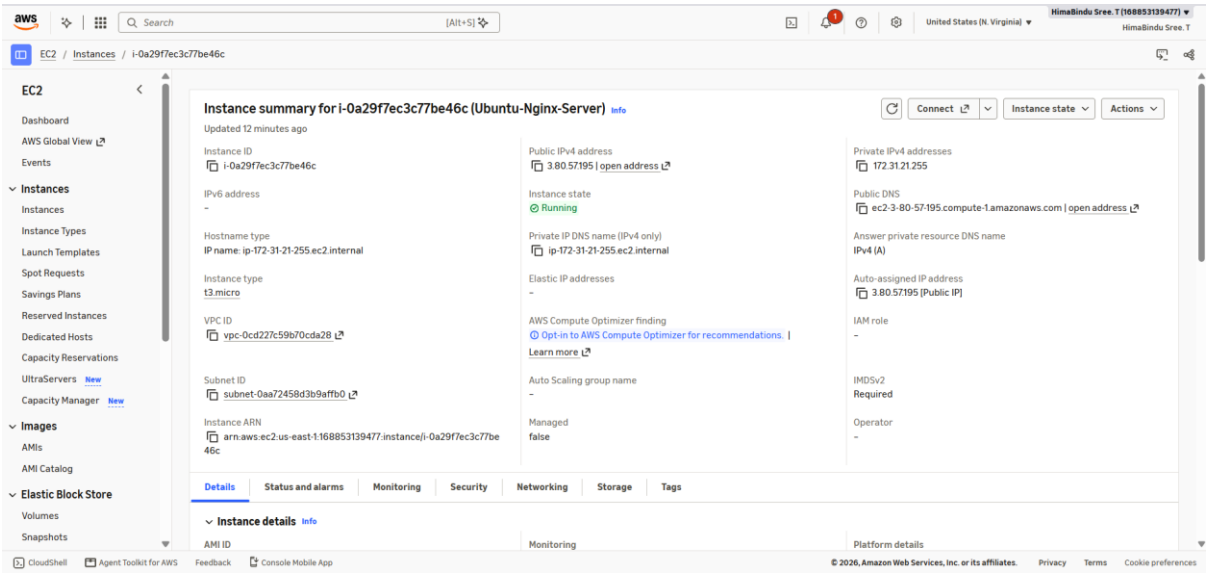
# Install Nginx web server
sudo apt install nginx -y

# Enable Nginx to start automatically on boot
sudo systemctl enable nginx

# Start the Nginx web server
sudo systemctl start nginx
```

3. Rechecked that the script was entered correctly, then clicked Launch instance.
4. Waited until the state showed Running and the status checks showed 2/2 checks passed.

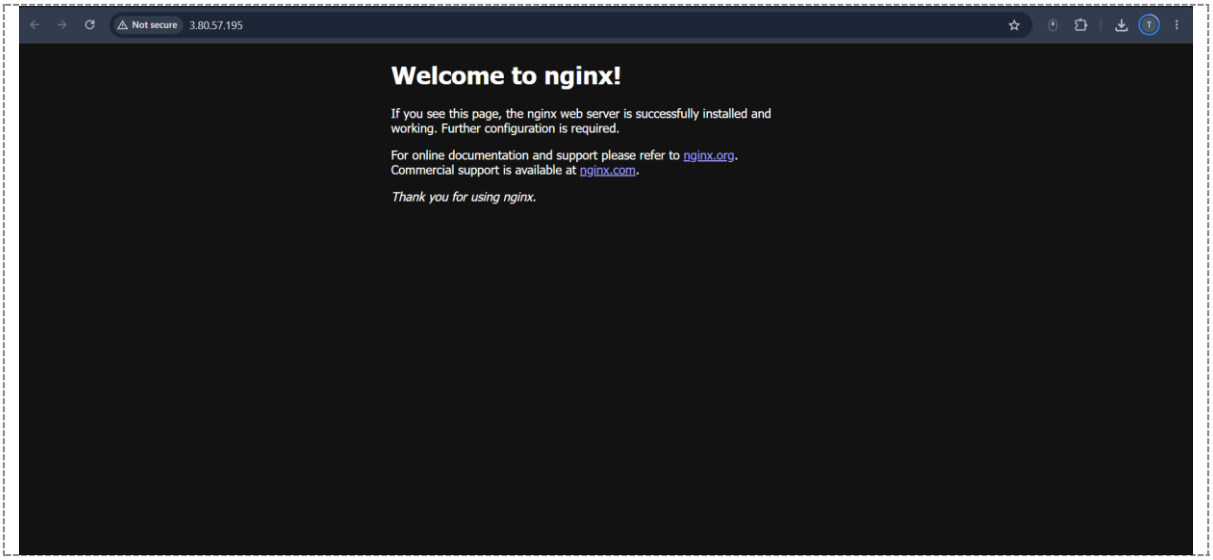
User data runs automatically on the first boot only, and the installation takes a few minutes, so I gave it some time before testing.



Screenshot 3: Instance running with 2/2 checks passed

Part C: Opening the Nginx default page

1. Selected the Ubuntu-Nginx-Server instance on the dashboard and copied its Public IPv4 address.
2. Opened a new browser tab and entered http://<Public-IPv4-address> in the address bar.
3. The default "Welcome to nginx!" page loaded, which confirmed the script worked.



Screenshot 4: Nginx welcome page in the browser with the IP in the address bar

What the User Data Script Does

Command	Purpose
apt update -y	Refreshes the package list
apt install nginx -y	Installs Nginx without asking for confirmation
systemctl enable nginx	Makes Nginx start on every boot
systemctl start nginx	Starts Nginx right now

Differences from Amazon Linux + Apache

Point	Amazon Linux + Apache	Ubuntu + Nginx
Package manager	yum / dnf	apt
Package name	httpd	nginx
Service name	httpd	nginx
Default login user	ec2-user	ubuntu
Config location	/etc/httpd/	/etc/nginx/
Default web root	/var/www/html	/var/www/html
Default page	Apache test page	"Welcome to nginx!" page
Repo refresh step	Not needed in the script	apt update needed first

The overall flow (launch, open port 80, user data, browser check) stayed the same. Only the OS-level commands and names changed.

Outcomes Achieved

- Launched an Ubuntu EC2 instance successfully
- Opened HTTP in the security group
- Understood that EC2 user data runs setup commands automatically at first boot
- Installed and started Nginx with a shell script, no SSH needed

- Checked that the Nginx service was running
- Opened the default Nginx page through the browser

Result

An Ubuntu EC2 instance was launched with Nginx installed through user data, and the default Nginx page was reached using the instance's public IPv4 address.